360° Business Intelligence

A Complete SaaS Company Health Analysis

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Executive Summary

A mid-sized SaaS company is experiencing simultaneous operational crises across four core business functions: workforce stability, customer retention, revenue growth, and supply chain performance. This report presents the findings of a comprehensive Business Intelligence engagement spanning five analytical pillars, five real-world datasets, and six industry-standard tools. The analysis identifies the root causes of each crisis, quantifies their business impact, and delivers four data-backed strategic recommendations with projected outcomes.

Key Performance Indicators

The following metrics represent the most critical findings across all five pillars of analysis:

16.1% Employee Attrition Rate	**26.5%** Customer Churn Rate	**$139K** Revenue Lost / Month
54.8% Late Delivery Rate	**39.8%** Sales Rep Attrition	**~80%** ML Model Accuracy

Findings by Pillar

Pillar 1 - Workforce Analytics (IBM HR Attrition)

Analysis of 1,470 employee records revealed that attrition is not random - it is concentrated and predictable. The Sales department loses 20.6% of its workforce annually, nearly double the company average. Employees working overtime are three times more likely to resign than those who do not. A $2,000 monthly income gap exists between employees who leave and those who stay. A Decision Tree ML model predicted individual attrition with 82.9% accuracy, enabling proactive HR intervention.

Key finding:

- Overtime policy and salary equity in Sales are the primary drivers of attrition
- Employees aged 20–30 leave at a 26.2% rate - the highest of any age group
- Employees at the company 0–2 years churn at 29.8% - onboarding and early engagement are critical

Pillar 2 - Revenue & Sales Analytics (Superstore Sales)

Analysis of four years of sales data ($2.3M total revenue) showed strong overall growth of 40% from 2016 to 2018. However, performance is unevenly distributed. The West region ($0.71M) outperforms the South region ($0.39M) by 82%. Technology generates the highest profit margins despite not being the top revenue category. The Furniture category shows consistently low profit margins despite significant sales volume.

Key finding:

- Technology category delivers the highest return on sales - the most efficient use of sales resources
- Furniture discounting is eroding margins without proportional volume gains
- Q4 consistently peaks - seasonality should drive inventory and staffing planning

Pillar 3 - Customer Churn Analysis (Telco Churn)

Of 7,043 customers analyzed, 1,869 (26.5%) had churned - representing $139,000 in lost monthly recurring revenue per month. Contract type is the single most powerful predictor of churn: month-to-month customers churn at 42.7%, compared to 11.3% for two-year contract holders. A Logistic Regression model achieved approximately 80% accuracy in identifying at-risk customers before they churned, enabling targeted retention campaigns.

Key finding:

- Contract structure - not product quality - is the primary churn driver
- Month-to-month customers who have been with the company less than 12 months are at highest risk
- Electronic check payment users churn at significantly higher rates - a billing experience indicator

Pillar 4 - Product & E-Commerce Analytics (Olist)

Analysis of 180,519 orders from Brazil's largest e-commerce marketplace revealed that Health & Beauty is the top revenue-generating category. The average customer review score across all orders is 4.1 out of 5.0. Python EDA identified and corrected 610 missing product category names - without this cleaning step, 8% of all category-level analysis would have been incorrect. São Paulo accounts for the highest order volume, making it the critical logistics hub.

Key finding:

- Data quality issues in raw data required Python intervention before any analysis was valid
- Health & Beauty and Computers & Accessories are the highest-value categories by revenue
- 73% of customers pay by credit card - payment infrastructure investment should prioritize this channel

Pillar 5 - Operational Efficiency (DataCo Supply Chain)

Analysis of 180,519 supply chain records identified a critical operational failure: 54.83% of all orders arrive late. The most counterintuitive finding is that First Class shipping - the most expensive and fastest-advertised tier - has a 95% late delivery rate. Same Day shipping, by contrast, achieves only 12% late deliveries. This is not a capacity issue; it is a routing, vendor, or process failure that requires immediate investigation.

Key finding:

- First Class shipping is performing worse than every other tier - customers are paying more for worse service
- Standard Class handles the highest volume (42K advance shipments) and performs adequately
- Computers generate the highest average profit per order ($158) - protecting this category's fulfilment is critical

Strategic Recommendations

The following four recommendations are directly derived from the data findings and are prioritised by estimated business impact. Each recommendation includes a specific, measurable outcome target.

#	Area	Action	Expected Impact
01	HR - IMMEDIATE Workforce Stability	Conduct a targeted salary review for Sales department employees and introduce an overtime reduction policy. The $2K/month income gap is the most actionable lever identified in this analysis.	Reduce Sales attrition from 20.6% to below 12% within 12 months. Estimated saving of over $50,000 per year in recruitment and training costs for every employee retained.
02	CX - HIGH IMPACT Revenue Recovery	Launch an annual contract conversion campaign targeting month-to-month customers identified as high-risk by the ML churn model. Offer a 10% discount for annual upgrades.	Recover $80,000-$100,000 in monthly recurring revenue. Reducing churn from 42.7% to 20% in this segment alone would recover over $50K/month.
03	OPS - URGENT Service Quality	Initiate an immediate root cause audit of First-Class shipping. Engage the logistics vendor for SLA review. Suspend upselling of First Class until the 95% late rate is resolved.	Eliminate the primary source of high-value customer complaints. Estimated reduction in refunds and customer service costs: $20,000-$40,000/month.
04	SALES - GROWTH Profit Optimization	Reallocate 20% of Furniture marketing budget to Technology category and West region campaigns. Introduce minimum margin thresholds before approving Furniture discount requests.	Projected 12-15% improvement in overall profit margin. West and Technology already lead in performance - this amplifies what is already working.

Methodology & Tools

Tool	Application	Deliverable
SQL (SQLite)	Data extraction & KPI queries	20+ queries across all 5 datasets - attrition rates, churn metrics, revenue trends, late delivery analysis
Python	Data cleaning, EDA & ML modelling	Day5_Data_Cleaning_EDA.py: 610 records fixed, 15 EDA charts. Day6_ML_Models.py: Logistic Regression (80% accuracy), Decision Tree (82.9% accuracy)

Power BI	Interactive dashboards	5 dashboards: Workforce, Revenue, Churn, Operations, Executive Summary - all with DAX calculated measures
Tableau	Visual analytics	4 interactive charts + 1 published dashboard on Tableau Public (public.tableau.com/app/profile/shailanshi.gupta/viz/360BIAttritionAnalysis)
PowerPoint	Executive presentation	12-slide deck with embedded dashboard screenshots, KPI summary, pillar findings, and strategic recommendations
Word	Executive report	This document - structured summary for leadership

Conclusion

> *The four crises facing this SaaS company are not independent. High attrition in Sales leads to knowledge gaps and weaker customer relationships. Weaker customer relationships drive churn. Churn drives revenue loss. Revenue pressure leads to cost-cutting that worsens working conditions and drives more attrition. Breaking this cycle requires addressing workforce stability first - it is the root cause that amplifies every other problem.*

The 360° BI project demonstrates that Business Intelligence - when applied systematically across every function of a business - does not just report on problems. It diagnoses their causes, quantifies their cost, and prescribes solutions with measurable outcomes. All project files, SQL queries, Python scripts, and Power BI dashboards are available on GitHub.

GitHub: github.com/ShailanshiGupta/360-BI-SaaS-Analysis ·

Tableau: public.tableau.com/app/profile/shailanshi.gupta/viz/360BIAttritionAnalysis